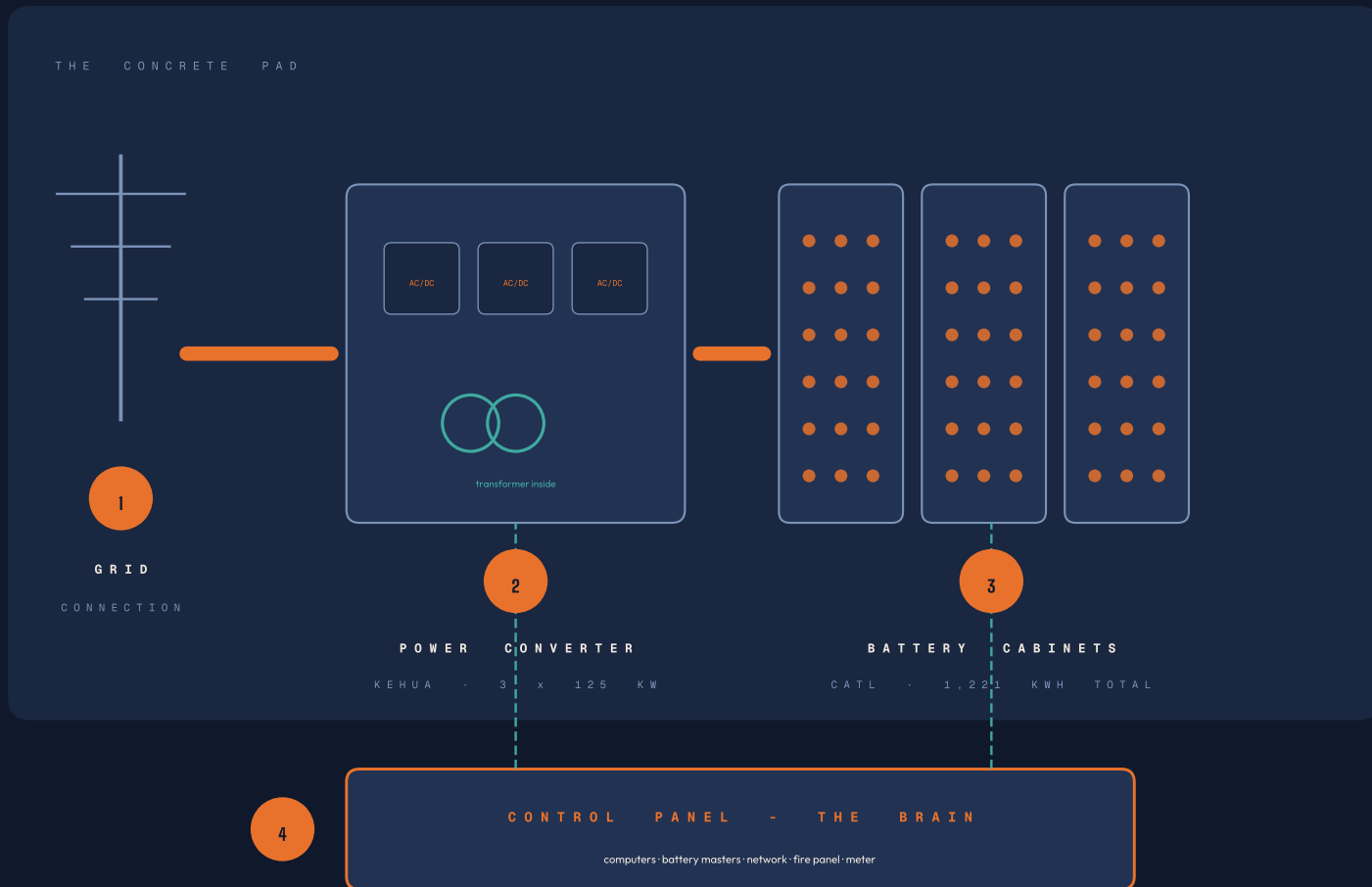


THE SYSTEM, ON ONE PAGE

Four pieces of equipment. One job: store energy, return it when Texas needs it.



1

GRID CONNECTION

480-volt line to ERCOT. Energy flows both ways.

2

POWER CONVERTER

Translates grid AC into battery DC and back - in both directions, in milliseconds. Its built-in transformer also feeds the cabinets' cooling.

3

BATTERY CABINETS

Three liquid-cooled cabinets of lithium cells. Each has its own cooling and its own fire suppression inside.

4

CONTROL PANEL

Built by STR Medical. Watches everything, sets the safe limits, talks to the outside world, and accounts for every kWh.

ONE CYCLE, STEP BY STEP

What actually happens, from a price signal to money in the meter.

01

L I S T E N

The site computers receive grid prices and dispatch signals from ERCOT, around the clock.

EVERY SECOND

02

A S K T H E B A T T E R I E S

Before acting, the computers ask the three battery masters: how full are you, how warm, how hard may we push? The masters answer every second.

03

D E C I D E

Cheap power on the grid? Charge. Grid stressed and paying? Discharge. The decision takes less than a second.

< 1 SECOND

04

C O N V E R T

The order goes to the power converter, which turns grid AC into battery DC (charging) or battery DC into grid AC (discharging).

05

P R O T E C T W H I L E M O V I N G

All through the cycle, thousands of cell sensors keep watch. If anything drifts - too warm, too full - the batteries slow down or disconnect themselves.

06

C O U N T I T

The meter records every kilowatt-hour in and out, on every phase. That record is the invoice.

This loop repeats all day, every day. Most of the time nothing dramatic happens -

which is exactly the point.

THREE KINDS OF POWER

The big power earns the money. Two small ones keep everything alive. Keeping them separate is the design.

THE BIG POWER

GRID < - > BATTERIES

Up to 375 kW flowing between grid and cells,

through the converter. This is the product.

It never passes through the control panel.

375 kW

THE HOTEL POWER

220 V FOR THE CABINETS

Cooling and cabinet electronics run on 220 V

from the transformer built into the converter

cabinet - included by Kehua, one breaker per

battery cabinet.

3 kW ea

THE LIFELOOD

24 V FOR THE BRAIN

The control panel makes its own 24 V DC for the

battery masters, network and monitoring - with

battery backup. If everything else fails, the

brain stays awake and calls for help.



WHY THE SEPARATION MATTERS

If HOTEL POWER fails

Cabinets lose cooling and pause - but the brain sees it instantly and reports. Nothing is damaged; the site waits.

If GRID POWER fails

No energy moves, but the brain rides through on battery backup, keeps watching, and restarts the site in order when power returns.

If the BRAIN loses a computer

There are two. The second one carries on alone while the first is replaced.

SAFE, SEEN, AND PROVEN

What happens on a bad day – and why we already know the answer.

A CELL RUNS WARM

Its sensors flag it within a second. That rack slows down; if it keeps warming, the rack disconnects itself. The other racks keep working.

SMOKE IS DETECTED

Siren and strobe fire. The cabinet disconnects and floods itself with fire-suppressing aerosol – no water involved. The fire panel runs on its own batteries throughout. A guarded abort button gives a trained person the final say.

THE INTERNET DROPS

On-site controllers keep running the plant safely on local control; monitoring resumes when the link returns.

THE GRID BLACKS OUT

Energy stops moving; the brain stays awake on battery backup, logs everything, and brings the site back step by step when the grid returns.

WHY WE ALREADY KNOW

- The design is a refinement of panels already running in the field in Carson, California.
- Every wire was checked against the battery and inverter makers' own factory documents.
- Automated tests re-verify all 148 circuits after any change – one wrong wire fails the build.
- An independent multi-specialist review challenged the design; everything it found is fixed.
- Final release still requires a licensed Professional Engineer's review and stamp.